Curriculum Vitae

Xenia Miscouridou

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Research Interests: Bayesian nonparametric statistics, modelling of network (graph) data, point processes, machine learning with an emphasis in explainability and interpretability

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Academic Positions

- 2023 – present, Lecturer in Statistics and Machine Learning (tenure-track),
  Department of Mathematics and Statistics, University of Cyprus
  (Approved maternity leave, 4 November 2024 - 6 April 2025.)
  - 2023 (August)–2026 (August), Honorary Lecturer, Department of Mathematics, Imperial College London, U.K.
- 2022–2023, Lecturer, Department of Mathematics and I-X Centre for AI in Science, Imperial College London, U.K.
- 2022, Postdoctoral Researcher and Junior Research Fellow, University of Oxford, U.K.
- 2021–2022, Postdoctoral Researcher in Statistics, Imperial College London, U.K.
- 2019, Visiting Researcher, Alan Turing Institute (National Institute of Data Science), U.K.
- 2018, Visiting Researcher, Courant Institute, New York University, U.S.A.

Industry Experience

- 2020, Machine Learning Scientist, Investments Artificial Intelligence Division, AIG, U.K.
- 2018, Data Science, Google, Zurich, Switzerland

University Education

- 2020, DPhil in Statistics (no corrections)
  University of Oxford, U.K.
  - Supervisors: François Caron and Yee Whye Teh.
- 2014–2015, MASt in Mathematics (Part III of the Mathematical Tripos),
  (Distinction)
  University of Cambridge, U.K.
  - Churchill Scholar for academic excellence
  - Essay: *Theoretical and Empirical Comparison of Bayesian and Frequentist Inference and Methodology for fMRI data analysis* (Distinction)
  - Essay supervisor: Professor John Aston
- 2011–2014, BSc in Mathematics (First Class)
  Imperial College London, U.K.

Grants and Funding

- ***DyNeMo:*** *Statistical Machine Learning for Dynamical Network Modelling*, European Commission, Marie Skłodowska-Curie Fellowship (Horizon Europe). €164,328. *2024–2028*
- ***Bayesian Nonparametrics for Sparse Graph Modelling***, Principal Investigator, University of Cyprus research start-up grant, €21,000. *2024–2027*

PhD students

- Marta Aparicio Rodriguez, jointly supervised with Dr Anastasia Borovykh and Prof Greg Pavliotis, Department of Mathematics, Imperial College London *2022–present*
- Wenqing Liu, jointly supervised with Dr Déborah Sulem, Faculty of Informatics, Università della Svizzera italiana *2025–present*

Research Assistants

- Marios Kyriacou, funded by my start-up grant *Bayesian Nonparametrics for Sparse Graph Modelling* *2024–2026*

MSc student supervision

- Co-advisor of Andria Ioannou's project, *Bayesian Neural Networks*, University of Cyprus *2025*
- Advisor of Iosif Pintirishis's project, *Detecting overlapping communities in U.S. equity markets: A Bayesian network model approach*, University of Cyprus *2024*
- Co-advisor of Federico Baio's thesis, *Semiparametric and Nonparametric Estimation of Multivariate Spatio-temporal Hawkes Processes*, Faculty of Informatics, Università della Svizzera italiana *2026*
- Co-advisor of Wenqing Liu's thesis, *Flexible Nonparametric Estimation of Spatio-temporal Hawkes Processes*, Faculty of Informatics, Università della Svizzera italiana *2025*
- Advisor of three theses on topics of Stochastic Processes and Gaussian Processes, MSc in Statistics, Department of Mathematics, Imperial College London *2021–2023*

Undergraduate student supervision

- Advisor of Antreas Laos, final-year thesis, *Dynamic Overlapping Community Detection on Sparse Networks*, University of Cyprus *2025–2026*
- Stanimira Georgieva, Research Project (non-ECTS), *An Overview of Graph Neural Networks for Social Network Analysis*, University of Cyprus *2026*

Publications

Preprints & Papers Under Review

- X. Miscouridou and D. Sulem, *Posterior concentration in spatio-temporal Hawkes processes*, arXiv:2601.03719v1, under revision at EJS, (2026).
- X. Miscouridou, D. Sulem and W. Liu, *Amortized Variational Inference of Latent Intensities from Replicated Point Patterns*, arXiv:2610.16669v1, (2026).
- W. Liu, X. Miscouridou and D. Sulem, *Flexible and Scalable Bayesian Modelling of Spatio-Temporal Hawkes Processes*, arXiv:2603.28556v1, (2026).
- X. Miscouridou, F. Panero and A. Laos, *Dynamic Sparse Graphs with Overlapping Communities*, arXiv:2512.10717v2, (2025).

Peer-reviewed Journal Articles

- Isaac Manring, Honglang Wang, George Mohler, Xenia Miscouridou, *BSTPP: a Python package for Bayesian spatiotemporal point processes*, Journal of Applied Statistics, 52(13), 2524–2543 (2025).
- X. Miscouridou, S. Bhatt, G. Mohler, S. Flaxman, S. Mishra, *Doubly stochastic spatiotemporal Poisson processes*, Transactions of Machine Learning Research[1], https://openreview.net/forum?id=xzCDD9i4IZ, (2023).
- M. Pakkanen, X. Miscouridou, M. J. Penn, C. Whittaker, T. Berah, S. Mishra, T. A. Mellan and S. Bhatt, *Unifying incidence and prevalence under a time-varying general branching process*, Journal of Mathematical Biology, 87(35) (2023).
- J. Lee, X. Miscouridou, F. Caron, *A unified framework for series representations and finite approximations of completely random measures*, *Bernoulli*, 29 (2022), 2142–2166.
- A. Brizzi, C. Whittaker, L.M.S Servo, et al., *Spatial and temporal fluctuations in COVID-19 fatality rates in Brazilian hospitals. Nature Medicine*, 28, 1509 (2022).
- N. R. Faria et al., *Genomics and epidemiology of a novel SARS-CoV-2 lineage in Manaus, Brazil*, *Science*, 372, 815–821 (2021).
- S. Mishra et al., *Changing composition of SARS-CoV-2 lineages and rise of Delta variant in England*, *eClinicalMedicine*, 39, article 101064 (2021).
- A. Todeschini, X. Miscouridou, F. Caron, *Exchangeable random measures for sparse networks with overlapping communities*, Journal of the Royal Statistical Society: Series B (Statistical Methodology), 82, 1–35 (2020).

Peer-reviewed Conference Proceedings

- M. Aparicio Rodriguez, X. Miscouridou, A. Borovykh, *Concept Reachability in Diffusion Models: Beyond Dataset Constraints*, Proceedings of the 42nd International Conference on Machine Learning (ICML), (2025).
- A. Hills, T. Tseriotou, X. Miscouridou, A. Tsakalides, M. Liakata, *Exciting Mood Changes: A Time-aware Hierarchical Transformer for Predicting Moments of Change in Longitudinally Segmented Social Media Posts*, Findings of the Association for Computational Linguistics (ACL), 12526–12537 (2024).
- I. Hawryluk, H. Hoeltgebaum, S. Mishra, X. Miscouridou, R. P. Schnekenberg, C. Whittaker, M. Vollmer, S. Flaxman, S. Bhatt, and T. A. Mellan, *Gaussian Process Nowcasting: Application to COVID-19 Mortality Reporting*, in *Proceedings of the Thirty-Seventh Conference on Uncertainty in Artificial Intelligence (UAI)*, PMLR, 161, 1258–1268 (2021) .
- X. Miscouridou, F. Caron, Y.W. Teh, *Modelling sparsity, heterogeneity, reciprocity and community structure in temporal interaction data*, Advances in Neural Information Processing Systems, (NeurIPS 2018), 2349–2358 (2018).
- X. Miscouridou, A. Perotte, N. Elhadad, R. Ranganath, *Deep Survival Analysis: Nonparametrics and Missingness*, Machine Learning for Healthcare, PMLR, 85, 1–12 (2018).

Workshop Proceedings

- Anthony Hills, Mahmud Elahi Akhter, Junyu Mao, Iqra Ali, Xenia Miscouridou and Maria Liakata, *Multistream Modelling for Mental Health: Modelling Linguistic and Temporal Contexts with Mutual and Self-Excitation in Social Media*, Workshop on Computational Linguistics and Clinical Psychology (CLPsych), ACL (2026).
- M. Willetts, X. Miscouridou, C. Holmes, S. Roberts, *Relaxed-Responsibility Hierarchical Discrete VAEs*, Bayesian Deep Learning Workshop, NeurIPS, (2021).

[1]TMLR is part of the Journal of Machine Learning Research Inc. organization and it is an electronic-only journal

Discussion Papers & Reports

- S. Mishra, S. Mindermann, M. Sharma, C. Whittaker, T. A. Mellan, T. Wilton, D. Klapsa, R. Mate, M. Fritzsche, M. Zambon, J. Ahuja, A. Howes, X. Miscouridou, G. P. Nason, O. Ratmann, G. Leech, J. F. Sandkühler, C. Rogers-Smith, M. Vollmer, H. J. T. Unwin, COVID-19 Genomics UK (COG-UK) Consortium, Y. Gal, M. Chand, A. Gandy, J. Martin, E. Volz, N. M. Ferguson, S. Bhatt, J. M. Brauner, and S. Flaxman, *Report 44: Recent trends in SARS-CoV-2 variants of concern in England*, Imperial College London COVID-19 Response Team, https://www.imperial.ac.uk/media/imperial-college/medicine/mrc-gida/20-05-20-COVID19-Report-44.pdf (2021).
- X. Miscouridou, F. Panero, *A discussion on Evaluating Sensitivity to the Stick-Breaking Prior in Bayesian Nonparametrics by R. Giordano, R. Liu, M. I. Jordan, T. Broderick.*, Bayesian Analysis, 18, 287–366 (2023).

Theses

- X. Miscouridou, *Bayesian Nonparametrics methods and applications in statistical network modelling*, DPhil dissertation, University of Oxford (2020).
- X. Miscouridou, *Theoretical and Empirical Comparison of Bayesian and Frequentist Inference and Methodology for fMRI data analysis*, (Essay for Part III of the Mathematical Tripos), University of Cambridge (2015).

Software

- A. Todeschini, X. Miscouridou, F. Caron, *SNetOC*, A MATLAB package for Exchangeable random measures for sparse networks with overlapping communities, GitHub repository https://github.com/misxenia/SNetOC (2020).
- Web-based interactive application (in progress and in collaboration with Antonis Ioannou) for generating and analysing random graphs https://app.misxenia.com/ (2026)

Scholarships and Awards

- 2023: Blackwell–Rosenbluth Award for Young Bayesian Statisticians, j-ISBA, junior section of ISBA (International Society of Bayesian Analysis)
- 2022: Forbes 30 Under 30 Europe, Science and Healthcare.
- 2019: Studentship award for a research appointment at the Alan Turing Institute
- 2016 & 2017: Scholarships for PhD studies from the Leventis Foundation, Cyprus
- 2015–2019: PhD studentship through EPSRC funding for OxWaSP Doctoral Training Center
- 2015: Award for academic excellence from Churchill College (Churchill Scholar list) in Part III of the Mathematical Tripos, University of Cambridge.
- Awards for academic excellence as an undergraduate student at the Department of Mathematics, Imperial College London
  - Gloucester Research Prize, third year Mathematics, 2013/14
  - Ken Allen Prize, first year Mathematics, 2011/12

Posters

- 2018: The Thirty-Second Annual Conference on Neural Information Processing Systems (NeurIPS)
- 2017: 10th International Workshop on Bayesian Inference in Stochastic Processes - BISP10 (Milan), Amazon Machine Learning Workshop (Berlin), Women in Machine Learning Workshop, NeurIPS conference (Long Beach)

Talks (invited in bold)

- 2027: **15th Conference on Bayesian Nonparametrics (Bergamo)**
- 2026: **35th European Meeting of Statisticians (Lugano), Statistics Seminar (UCL)**, **Computational Science Symposium CONVERGE (University of Cyprus)**
- 2025: **IMS Conference on Statistics and Data Science (ICSDS) (Seville), Discussant at Objective Bayes Conference (Athens)**
- 2024: **Royal Statistical Society International Conference (Brighton)**, ISBA World Meeting (Venice), **Keynote Speaker at the Workshop in Statistics (Athens University of Economics and Business)**
- 2023: IMS Conference on Statistics and Data Science (ICSDS) (Lisbon), **EcoSta (Tokyo), CIRM conference on Approximation Methods in Bayesian Analysis (Marseille), *Keynote Speaker* at Pancyprian Statistical Conference (Cyprus), Statistics Seminar (University of Lancaster), Webinar of ISBA-Bayesian Nonparametrics section (Zoom), Workshop on Bayesian Inference in Stochastic Processes, BISP13 (Madrid), TEDx Speaker on Memory and AI at National Technical University of Athens (Athens)**
- 2022: 13th Conference on Bayesian Nonparametrics (Chile), **Statistics seminar (Imperial College London), Statistics and Econometrics seminar (University of Tokyo), Statistics seminar (University College London)**, **International Symposium for Nonparametric Statistics - ISNPS (Cyprus), ISBA Bayesian Nonparametrics Networking Workshop (Cyprus).**
- 2021: **Data Science Seminar Series (University of Cyprus), seminars of Women in Machine Learning and Data Science - WiMLDS (Joint Paris and Limassol meetup)**
- 2020: **Bayesian Young Statisticians Meeting (BAYSM) (j-ISBA)**
- 2019: **Semi and Nonparametric Modelling, CMStatistics (London)**, 12th Conference on Bayesian Nonparametrics (Oxford), **Mathematical Institute Seminar (University of Oxford)**
- 2018: Greek Stochastics κ' on Statistical Learning (Athens)
- 2017: 11th Conference on Bayesian Nonparametrics (Paris)

Outreach

A. Seminars in Machine Learning Industry Research Groups

- *Statistical Machine Learning for real-world problems*, Google Data Science (Google Zurich), 2024
- *Spatiotemporal modelling with Hawkes Processes*, Amazon Research Center (Amazon Berlin), 2023
- *On Bayesian Machine Learning*, IBM Research (Boston MA), 2020
- *Bayesian Nonparametrics for Complex Networks*, Microsoft Research, (Cambridge U.K.), 2020

B. Seminars in Machine Learning Summer Schools

- *An overview of Variational Autoencoders and Generative Adversarial Networks*, Mediterranean Machine Learning Summer School, Anatolia College (Thessaloniki), 2023
- *Lecture on Clustering Methods*, and *Lecture on Dimensionality Reduction Methods*, Machine Learning and Applied Statistics, Imperial College London (London), Summer 2022 & 2023

Conference, workshop and seminar organizer

- Scientific Committee member, Bayesian Young Statisticians Meeting (BAYSM), 2027
- Scientific Committee member, ISBA Bayesian Nonparametrics networking workshop, Seoul, 2026
- Session organizer, CFE-CMStatistics, Berlin, 2026
- Scientific Committee member, Bayesian Nonparametrics Conference (BNP15), UCLA, 2025
- Organizing Committee member, Bayesian Young Statisticians Meeting (BAYSM), 2024
- Local Organizing Committee member, ISBA Bayesian Nonparametrics networking workshop, Cyprus, 2022
- Organizing Committee member, ISBA World Meeting, 2021
- Programme Chair (equivalent to area chair for scientific committee), Women in Machine Learning Workshop, co-located at NeurIPS, 2021.
- Co-organizer, Computational Statistics and Machine Learning reading group, Imperial College London, 2022

Teaching Experience

- Lecturer of *Probability and Statistics*, Winter terms 2023/24, 2024/25, 2025/26, 2026/27
  University of Cyprus (UG course on topics such as Probability, Random Variables, Central Limit Theorem, Hypothesis Testing, Simple Linear Regression)
- Lecturer of *Statistical Learning*, Easter terms 2023/24, 2025/26
  University of Cyprus (MSc Statistics course on statistical learning, linear regression, classification, model selection, regularization, models beyond linearity)
- Lecturer of *Unsupervised Learning*, MLDS MSc, Autumn term 2022/23,
  Imperial College London (MSc course on methods for unsupervised learning in machine learning)
- Class Tutor, *Foundations of Statistical Inference*, Winter terms 2017/18, Department of Statistics, University of Oxford.
- Teaching assistant of *Foundations of Statistical Inference*, Winter terms 2016/17
  University of Oxford
- Leading Tutor *Introductory Course in Mathematics*, September 2016 & 2017, residential and online teaching, St Peter's College, University of Oxford.
- Tutor in *Practical sessions in R language for statistical inference problems*, University of Oxford.

Service

- Associate Editor, *Journal of the Royal Statistical Society: Series C (Applied Statistics)* (2026–present)
- Referee for *Bayesian Analysis*, *Electronic Journal of Statistics*, *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, and *Statistics in Medicine*
- Referee for NeurIPS, ICML, AISTATS, UAI, and the Women in Machine Learning (WiML) Workshop
- Referee for grants submitted to UK Research and Innovation (UKRI)
- Chair, junior section of the International Society for Bayesian Analysis (j-ISBA) (2022)
- Chair-Elect, junior section of the International Society for Bayesian Analysis (j-ISBA) (2021)
- PhD admissions interviewer, Imperial College London & University of Oxford, StatML CDT (2021)
- UG admissions interviewer for the degree of Mathematics, University of Oxford, (2016)
- Mentoring of PhD students, University of Oxford (2016/17)